

- The Williamson’s nonlinear test case 2 [44], consisting of a geostrophically balanced steady solution, with mean geopotential $\bar{\Phi} = 2.94 \times 10^4 \text{m}^2/\text{s}^2$, mean advective velocity $|\bar{u}| \approx 39 \text{m/s}$ and characteristic speed $c \approx 210 \text{m/s}$. The authors recommend error evaluations after 5 days of integration; here, we execute simulations for 7 days.
- The geostrophic balance test with bathymetry proposed by [27], which is a modification of Williamson’s test case 2. In this test case, the fluid depth is constant ($h = h_0$) and the bathymetry profile is constructed to ensure geostrophic balance. By choosing a relatively small h_0 , nonlinear effects are not negligible, allowing us to evaluate the influence of the discretization of the nonlinear terms on the convergence order. We consider $h_0 = 100 \text{m}$ and $h_0 = 1 \text{m}$, with mean geopotential $\bar{\Phi} = gh_0$, mean advective velocity $|\bar{u}|$ and characteristic speed c respectively $(\bar{\Phi}, |\bar{u}|, c) \approx (981, 39, 70)$ and $(\bar{\Phi}, |\bar{u}|, c) \approx (9.81, 39, 42)$ (in m^2/s^2 , m/s and m/s). The simulations are also executed for 7 days.
- The unstable jet test case proposed by [12], which is a popular and challenging benchmark in atmospheric modeling. In this test, with null bathymetry and whose initial solution is a stationary zonal jet, a small Gaussian bump perturbation on the geopotential field leads to the formation of characteristic vortices and important nonlinear processes that transfers energy along the spectra, and allows enstrophy cascade to the small-scale components of the solution. The mean fluid depth and geopotential are respectively $\bar{h} = 10000 \text{m}$ and $\bar{\Phi} \approx 98100 \text{m}^2/\text{s}^2$, and the advective velocity is $|\bar{u}| = 80 \text{m/s}$, resulting in the characteristic speed $c \approx 393 \text{m/s}$. A similar test case has been used for numerical validation of SE11 and SE12 on the plane by [26] and, as done in that work, errors are measured after one day of simulation, before the formation of the vortices. We also conduct qualitative analyses considering longer times of simulation of this test case.

In all test cases and for all schemes, we perform simulations by keeping the ratio between the temporal and spectral resolutions constant, with $\Delta t \in \{60, 120, 240, 480, 960\}$ (s) and $M \in \{512, 256, 128, 64, 32\}$, such that the CFL numbers are kept constant for each test case (namely, 0.48, 0.16, 0.10 and 0.91, respectively for the Williamson’s test case 2, the geostrophically balanced test case with bathymetry ($h_0 = 100 \text{m}$ and $h_0 = 1 \text{m}$), and the unstable jet test case). The errors in each simulation are computed w.r.t. a reference solution integrated using a fourth-order explicit Runge-Kutta scheme, with a time step four times smaller and the same spectral resolution (thus allowing us to analyze exclusively the errors due to the temporal discretization without imposing too severe stability constraints by considering a fixed high spectral resolution). None of the simulations use artificial viscosity, *i.e.*, $\nu = 0$. Figure 5 reveals that, among the semi-Lagrangian exponential methods, the second-order accuracy is achieved only by the proposed SE22 scheme, with all the other semi-Lagrangian exponential schemes being of first order. This is observed in all test cases, including those with relatively important nonlinear effects (geostrophic balance with topography and small h_0). It shows that both our proposed modification of the linear term and the second-order discretization of the nonlinear one proposed by [26] are required to ensure global second-order accuracy.

Figure 5 also confirms our hypotheses on the stability and accuracy of semi-Lagrangian exponential methods. These schemes, notably those with a second-order discretization of the nonlinear term, have improved stability properties compared to the Eulerian ones; indeed, ETD2RK and mainly ETD1RK are unstable in most of the simulations. The only simulations in which ETD2RK is stable are those with less important nonlinear effects, meaning that a weaker enstrophy cascade takes place, namely the geostrophically balanced steady-state solution (Figure 5a) and the short-run unstable jet simulation (Figure 5d). In both simulations, ETD2RK and SE22 outperform

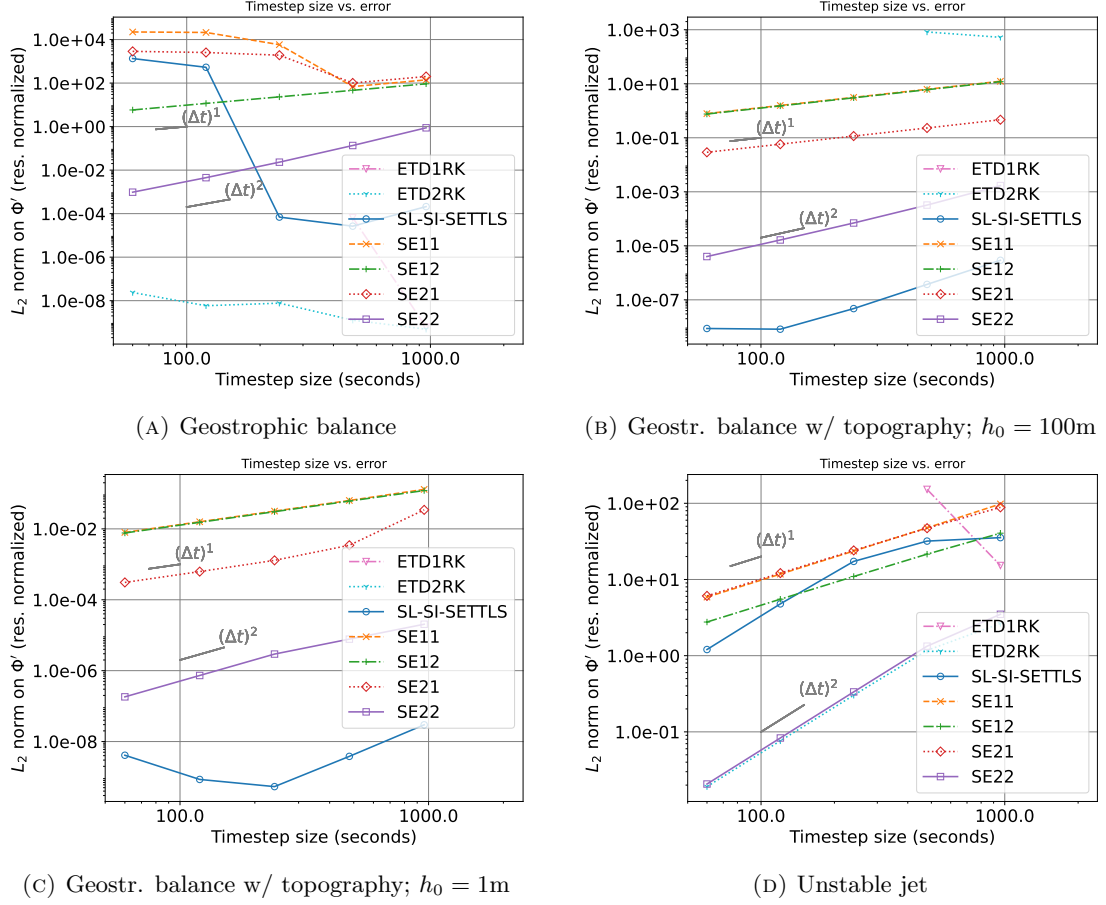


FIGURE 5. Numerical verification of the convergence order of the schemes considered in this work (normalized L^2 norm of the error on the geopotential field perturbation *vs* the time step size). Similar results are obtained with the L^∞ norm.

the accuracy of the also second-order accurate SL-SI-SETTLS (which develops instabilities in the geostrophic balance test case), reflecting the accurate integration of the linear term by the exponential schemes. Possibly, the observed instabilities in the geostrophic balance test case (which arise after 3-4 days of simulation) are due to the amplification of small departures from the steady-state solution, which may be caused by the time-splitting approach for integrating equations including linear and nonlinear terms; a detailed study on these amplifications is conducted in Section 7.3. On the other hand, SL-SI-SETTLS is more accurate in the geostrophic balance test case with topography (Figures 5b and 5c), in which the nonlinear effects are more relevant due to the small mean depth h_0 . In this case, although still convergent with the expected accuracy, the semi-Lagrangian exponential schemes produce larger errors. We remark, however, that the observed second-order accuracy of SE22 in this test case confirms the discussion in Section 4 in the sense that the source of

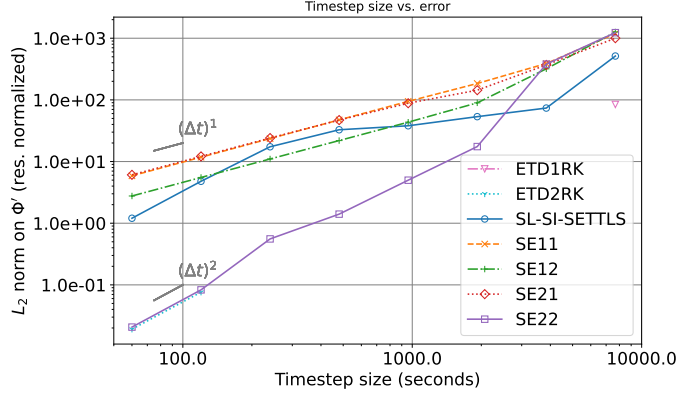


FIGURE 6. Numerical verification of the convergence order of the schemes considered in this work (normalized L^2 norm of the error on the geopotential field perturbation *vs* the time step size). All simulations consider the same spectral resolution ($M = 256$); the reference solution is computed using a fourth-order explicit Runge-Kutta scheme, with $M = 512$ and $\Delta t = 2$ s.

the first-order accuracy of SE12 is indeed the discretization of the linear term, and no modification of the discretization of the nonlinear term is required for improving its order of accuracy. Finally, note that some simulations reach an error plateau around $10^{-8} - 10^{-9}$, which is a consequence of our formulation considering the geopotential perturbation, $\Phi = \bar{\Phi} + \Phi'$, with $\bar{\Phi}$ being several orders of magnitude larger than Φ' ; therefore, we consider these solutions to have converged.

Since our focus in these first results was on the convergence properties of the schemes, we ensured stability in almost all simulations by considering a fixed CFL number. However, we now highlight the advantages of the proposed SE22 w.r.t. the other schemes. We repeat the simulations of the unstable jet test case, but now with a fixed spectral resolution ($M = 256$) and time step sizes chosen in $[60, 8192]$ (s), with approximate smallest and largest CFL numbers $c\Delta t/\Delta x$ of 0.45 and 62, respectively. The reference solution is computed using a fourth-order explicit Runge-Kutta scheme, with $M = 512$ and $\Delta t = 2$ s, resulting in an approximate CFL number of 0.03. The results are presented in Figure 6. It is clear that the proposed second-order semi-Lagrangian exponential scheme SE22 “extends” the error curve of the second-order Eulerian exponential method ETD2RK, *i.e.*, it has a largely improved stability range, maintaining approximately the same accuracy; indeed, the second-order accuracy of SE22 is observed with time step sizes as large as approximately 2000s, corresponding to CFL numbers as large as 15. (the small peak in the SE22 curve will be discussed in Sections 7.3 and 7.4), whereas the Eulerian scheme is unstable for $\Delta t > 120$ s. It indicates a better ability of the proposed method to stably propagate fast gravity waves. We also observe that, in this test case, SE22 is much more accurate than SL-SI-SETTLS and the first-order semi-Lagrangian exponential schemes.

7.3. Simulation-based stability analysis

As discussed in Section 6, the lack of commutation between the exponential and interpolation operators does not allow us to compare the stability properties of the semi-Lagrangian exponential